



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE (Advanced)
ALLEN COMMON TEST
14-08-2022

JEE(Main + Advanced) : Enthusiast Course [Phase : S, TRAS, I(A), I, II, III & LIVE- I(A), I, I(E)]

ANSWER KEY

PAPER-1

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	B	A	A	A		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	A,C	B	A,D	B,D	B,C	B,D
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	2.09	2.61 to 2.62	1.25	0.75	83.33	9.99 to 10.00
SECTION-II (ii)	Q.	7	8	9			
	A.	7	4	20			

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4		
	A.	C	D	B	D		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	B,C,D	A,B,D	B,D	B,C	A,B,C,D	A,B,D
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	10.00	10.00	5.00	3.00	60.00	213.75
SECTION-II (ii)	Q.	7	8	9			
	A.	18	3	4			

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	D	C	B	B		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	A,C	A,B,C	A,C,D	A,B	B,C	B,C
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	9.00	8.00	2.00	1.00	5	2
SECTION-II (ii)	Q.	7	8	9			
	A.	5	9	5			

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (B)

$$T = \pi \sqrt{\frac{m}{k}} + \pi \sqrt{\frac{m}{4k}} = \pi \left[\sqrt{\frac{4}{4}} + \sqrt{\frac{4}{16}} \right]$$

$$= \pi \left[1 + \frac{1}{2} \right] = \frac{3}{2} \pi$$

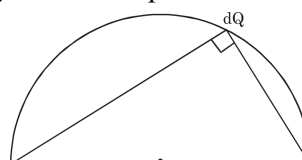
$$\frac{1}{2} k A^2 = \frac{1}{2} m v_{\max}^2$$

$$\frac{1}{4} \times 4 \times (2)^2 = \frac{1}{2} \times 4 v_{\max}^2$$

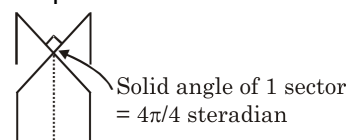
$$v_{\max} = 2 \text{ m/s}$$

2. Ans (A)

Consider elementary charged element along the length of the half cylinder. The top view is shown below :



The rectangular face of the half cylinder subtends a solid angle of $\frac{1}{4} \times 4\pi$ steradian at every linear dQ .

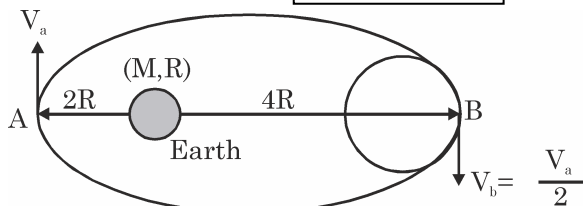


3. **Ans (A)**

$$\frac{X}{100 - X} = \frac{nr/2}{2r/n}$$

4. **Ans (A)**

$$\frac{1}{2} m V_a^2 - \frac{GMm}{2x} = \frac{1}{2} m \frac{V_a^2}{4} - \frac{GMm}{4x}, \quad V_a = \sqrt{\frac{2}{3} \frac{GM}{x}}$$



ROC at point - B

$$ROC = \frac{V_b^2}{a\ell} = \frac{\frac{2GM}{3R} \times \frac{1}{4}}{\frac{GM}{(4R)^2}} = \frac{8R}{3}$$

PART-1 : PHYSICS

SECTION-I (ii)

7. **Ans (A,D)**

From Q to B, KVL

$$V_Q - 4i - 6 - 2i = V_B \Rightarrow i = 3A$$

$$\text{From A to B KVL} \Rightarrow V_A - V_B = 48V$$

8. **Ans (B,D)**

$$C = \frac{\epsilon_0 A}{d_1}, C' = \frac{\epsilon_0 A}{d_2}$$

$$\text{Extra charge flown} = Q' - Q = (C' - C) V$$

$$= \epsilon_0 AV \left[\frac{1}{d_2} - \frac{1}{d_1} \right]$$

Work done by battery :

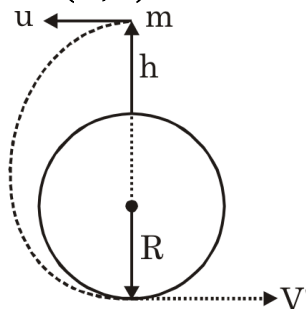
$$W_b = V \times \text{charge flown} = \epsilon_0 AV^2 \left[\frac{1}{d_2} - \frac{1}{d_1} \right]$$

Change in P.E. of capacitor

$$= \Delta U = \frac{1}{2} (C' - C) V^2$$

$$= \frac{1}{2} \epsilon_0 AV^2 \left[\frac{1}{d_2} - \frac{1}{d_1} \right]$$

9. **Ans (B,C)**



By AMC

$$mu(h + R) = mV'R$$

$$\frac{1}{2} mu^2 - \frac{GMm}{R+h} = \frac{1}{2} mV'^2 - \frac{GMm}{R}$$

10. **Ans (B,D)**

(A) Area covering and not covering the dielectrics are in the ratio $1/5 : 4/5 = 1:4$.

As both parts are at the same potential thus for equal charges on both, two parts shall have same capacitance. Thus, dielectric constant is 4.

(B) Charge induced on the face of the dielectric slab is

$$q_{\text{ind}} = 0.5Q \left(1 - \frac{1}{k} \right) = 0.5 \times 20 \times (1 - 0.25) = 7.5C$$

(C)/(D) Force of interaction between the two plates will not depend upon the dielectric but depends upon the charge density.

$$F = \frac{Q^2}{2\epsilon_0 A}$$

PART-1 : PHYSICS

SECTION-II (i)

1. **Ans (2.09)**

When the bob is displaced slightly by angle θ .

$$\tau = mg \sin \theta \ell + kx\ell = mg \sin \theta \ell + k\ell^2 \theta \quad (x = \ell \theta)$$

$$m\ell^2 \frac{d^2 \theta}{dt^2} = -(mg\ell + k\ell^2) \theta$$

$$\frac{d^2 \theta}{dt^2} = - \left(\frac{g}{\ell} + \frac{k}{m} \right) \theta$$

$$\omega = \sqrt{\left(\frac{g}{\ell} + \frac{k}{m} \right)} = 3$$

$$T = \frac{2\pi}{\omega}$$

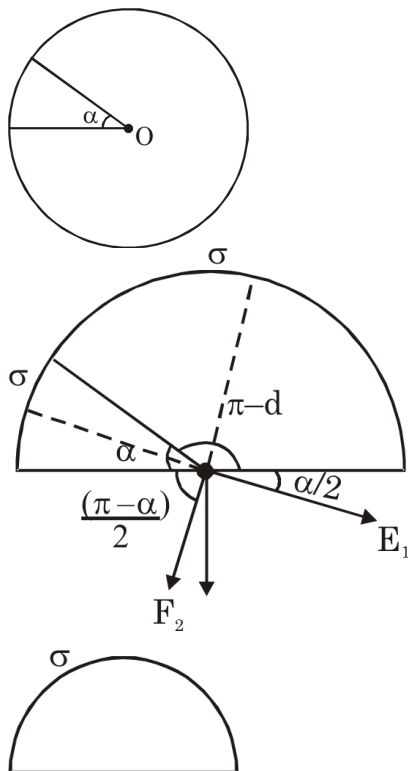
2. Ans (2.61 to 2.62)

In this case the cord is similar to spring when it is stretched but it will exert no force during half the motion when it is loose.

$$T = \frac{T_1}{2} + \frac{T_2}{2} = \frac{\pi}{\omega_1} + \pi \sqrt{\frac{L}{g}}$$

$$= \frac{\pi}{3} + \frac{\pi}{2} = \pi \left(\frac{5}{6} \right) = 2.62$$

4. Ans (0.75)



$$\frac{\sigma}{4\epsilon_0} = E_1 \sin \frac{\alpha}{2} + E_2 \cos \frac{\alpha}{2} \quad \dots (1)$$

$$E_1 \cos \frac{\alpha}{2} = E_2 \sin \frac{\alpha}{2} \quad \dots (2)$$

$$\frac{\sigma}{4\epsilon_0} = E_1 \sin \frac{\alpha}{2} + \left[\frac{E_1 \cos \frac{\alpha}{2}}{\sin \frac{\alpha}{2}} \right] \cos \left(\frac{\alpha}{2} \right)$$

$$= \frac{E_1}{\sin \frac{\alpha}{2}}$$

$$E_1 = \frac{\sigma}{4\epsilon_0} \sin \frac{\alpha}{2}$$

$$\text{For } \sigma = 5\epsilon_0 \text{ C/m}^2 \text{ and } \alpha = 2\cos^{-1}(4/5),$$

$$E_1 = 0.75 \text{ N/C}$$

6. Ans (9.99 to 10.00)

$$G \begin{cases} S_{I_g} = 10 = \frac{N}{I_g} = \frac{150}{I_g} \Rightarrow I_g = 15\text{mA} \\ S_{V_g} = 2 = \frac{N}{V_g} = \frac{150}{V_g} \Rightarrow V_g = 75\text{mV} \end{cases}$$

$$G = \frac{V_g}{I_g} = \frac{75}{15} = 5\Omega$$

$$AM \Rightarrow I = \frac{6A}{\text{divn}} \times 150 \text{ divn.} \Rightarrow I = 900A, S = \frac{I_g \cdot G}{(I - I_g)}$$

$$S = \frac{(15 \times 10^{-3}) \times 5}{900 - 15 \times 10^{-3}} = 8.3 \times 10^{-5} \Omega \Rightarrow \boxed{83.33} \mu\Omega$$

$$VM \Rightarrow V = \frac{1V}{\text{div.}} \times 150 \text{ div.} \Rightarrow V = 150V, R = G \left(\frac{V}{V_g} - 1 \right)$$

$$R = G \left(\frac{150}{75 \times 10^{-3}} - 1 \right) = 9995\Omega$$

Ans. 9.99 to 10.00

PART-1 : PHYSICS

SECTION-II (ii)

7. Ans (7)

$$M = \frac{4}{3}\pi R^3 \rho_1 + \frac{4}{3}\pi \left[(2R)^3 - R^3 \right] \rho_2$$

$$M = \frac{4}{3}\pi R^3 [\rho_1 + 7\rho_2]$$

$$\text{At surface } g_1 = \frac{GM}{(2R)^2} = \frac{G}{4R^2} \times \frac{4}{3}\pi R^3 (\rho_1 + 7\rho_2)$$

At depth R

$$g_2 = \frac{GM}{R^2} = \frac{G}{R^2} \times \frac{4}{3}\pi R^3 \rho_1$$

$$= \frac{4\pi GR}{3} \rho_1$$

$$\boxed{g_1 = \frac{1}{2}g_2}$$

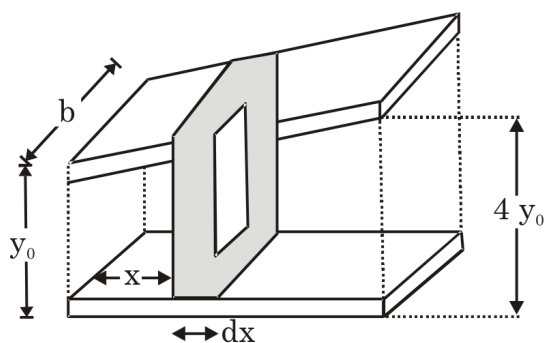
$$\rho_1 + 7\rho_2 = 2\rho_1$$

$$7\rho_2 = \rho_1$$

$$\Rightarrow \boxed{\frac{\rho_1}{\rho_2} = 7}$$

8. **Ans (4)**

We consider a differential strip of width dx and length b to approximate a differential capacitor of area $b \cdot dx$ and separation $d = y_0 + \left(\frac{3y_0}{a}\right)x$. All such differential capacitors are in parallel arrangement.



$$dC = \frac{\epsilon_0 (b dx)}{\left(y_0 + \frac{3y_0}{a}x\right)} \Rightarrow C = \int dC$$

$$C = \epsilon_0 b \int_0^a \frac{dx}{\left(y_0 + \frac{3y_0}{a}x\right)} = \frac{\epsilon_0 b a}{3y_0} \left[\ln \left(y_0 + \frac{3y_0 x}{a}\right) \right]_0^a$$

$$= \frac{\epsilon_0 b a}{3y_0} \ln(4)$$

Hence, $k = 4$

9. **Ans (20)**

Series and parallel combination

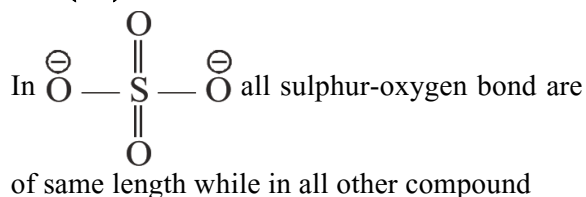
PART-2 : CHEMISTRY

SECTION-I (i)

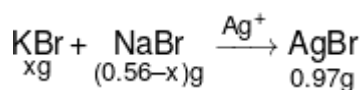
1. **Ans (C)**

Second electron gain enthalpy is always an endothermic process. For any element energy is required to add an electron in an anion.

2. **Ans (D)**



3. **Ans (B)**



Applying POAC on Br atoms

$$\frac{x}{119} + \frac{(0.56-x)}{103} = \frac{0.97}{188}$$

$$x = 0.1332 \text{ g}$$

4. **Ans (D)**

$$\text{Bond energy} \propto \frac{1}{\text{stability of free radical}}$$

PART-2 : CHEMISTRY

SECTION-I (ii)

5. **Ans (B,C,D)**

H_2O_2 protonates to a larger extent than H_2O in aqueous solution.

$\text{N}(\text{SiH}_3)_3$ attains sp^2 hybridisation. The lone pair of N is involved in $p\pi-d\pi$ back bonding.

7. **Ans (B,D)**

$A \rightarrow S_N1$ Mechanism

$B \rightarrow E_2$ Mechanism

$C \rightarrow E_1$ Mechanism

$D \rightarrow S_N2$ Mechanism

8. **Ans (B,C)**

Rate of S_N1 depends upon the stability of carbocation.

10. **Ans (A,B,D)**

$$\frac{d(\ln k)}{dT} = \frac{E_a}{RT^2} = \frac{0.04T^2}{2T^2} = 0.02$$

$$\ln\left(\frac{K_2}{K_1}\right) = 0.02 \times 200 = 4$$

$$\therefore \left(\frac{K_2}{K_1}\right) = e^4$$

PART-2 : CHEMISTRY

SECTION-II (i)

2. **Ans (10.00)**

In all form of X, Cl is outside the co-ordination sphere having formula $[\text{Pt}(\text{gly})(\text{en})\text{ClBr}]\text{Cl}$. In this case electron pair denoted by nitrogen and halogen is 5. Pale yellow compound having formula $[\text{Pt}(\text{gly})(\text{en})\text{Cl}_2\text{Br}]$. In this case electron pair donated by nitrogen and halogen is 5 sum of $a + b = 5 + 5 = 10$

3. **Ans (5.00)**

E_2 Mechanism

4. **Ans (3.00)**

a, b, g are correct

6. **Ans (213.75)**

Sucrose + $\text{H}_2\text{O} \rightarrow \text{Glucose} + \text{Fructose}$

0.0125 - -

0.0125-x x x

$\Delta T_b = m_1 K_b + m_2 K_b + m_3 K_b$

$m_1 + m_2 + m_3 = \frac{0.104}{0.52} = 0.2$

$\frac{0.0125 - x + x + x}{100} \times 1000 = 0.2$

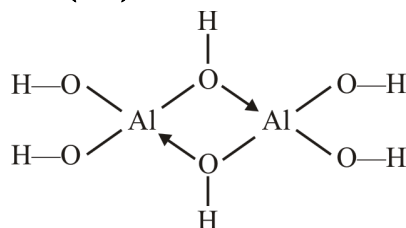
$x = 0.0075$

mol % of sucrose

PART-2 : CHEMISTRY

SECTION-II (ii)

7. **Ans (18)**



Total no. of $(2C - 2e)$ bond = $10 = x$.

Total no. of $(3C - 2e)$ bond = $y = 0$

Total no. of $(3C - 4e)$ bond = $z = 2$

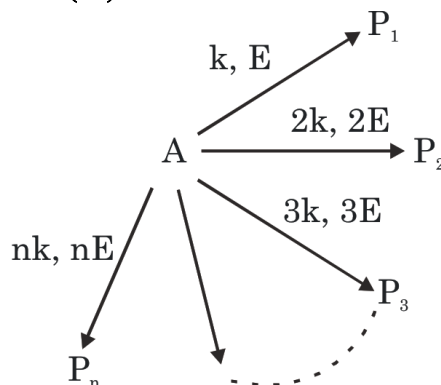
a = total no. of $(2C - 2e)$ bond = 4

b = total no. of $(3C - 2e)$ bond = 2

c = total no. of $(3C - 4e)$ bond = 0

Now, $x^c + b^y + z^a = 10^0 + 2^0 + 16 = 18.00$

8. **Ans (3)**



Let rate constant

$$k_0 = k + 2k + 3k + \dots + nk$$

$$k_0 = A e^{-\frac{E_0}{RT}}$$

$$\frac{dk_0}{dT} = A e^{-\frac{E_0}{RT}} \left(\frac{E_0}{RT^2} \right)$$

$$\frac{dk_0}{dT} = k_0 \frac{E_0}{RT^2}$$

$$\frac{1}{k_0} \left[\frac{d(k + 2k + \dots + nk)}{dT} \right] = \frac{E_0}{RT^2}$$

$$\frac{1}{k_0} \left[\frac{k \cdot E}{RT^2} + \frac{2k \cdot 2E}{RT^2} + \frac{3k \cdot 3E}{RT^2} + \dots + \frac{nk \cdot nE}{RT^2} \right] = \frac{E_0}{RT^2}$$

$$\frac{k \cdot E}{RT^2 k_0} [1 + 2^2 + 3^2 + \dots + n^2] = \frac{E_0}{RT^2}$$

$$\frac{k \cdot E}{RT^2 \cdot k_0} \frac{(n)(n+1)(2n+1)}{6} = \frac{E_0}{RT^2}$$

$$\frac{k \cdot E}{k(1+2+3+\dots+n)} \cdot \frac{(n)(n+1)(2n+1)}{6} = E_0$$

$$\frac{E n(n+1)(2n+1)}{6 \cdot \frac{n(n+1)}{2}} = E_0$$

$$E_0 = \frac{(2n+1)E}{3}$$

9. **Ans (4)**

1, 3, 4, 5

PART-3 : MATHEMATICS

SECTION-I (i)

1. **Ans (D)**

$$f(3) = f(5^{1/4}) = 0 \quad \text{Many one}$$

$$f(x) \neq -\sqrt{5} \quad \text{for any real value of } x.$$

so, $f(x) \rightarrow \text{Into}$

2. **Ans (C)**

$$\lim_{x \rightarrow \infty} \frac{x^2(a) + x + 1 - b^2x^2}{\sqrt{x^2(a) + x + 1} - bx} = \frac{1}{2}$$

$$\lim_{x \rightarrow \infty} \frac{x(1 + \frac{1}{x})}{x(\sqrt{a} + \frac{1}{x} + \frac{1}{x^2} - b)} = \frac{1}{2}$$

$$\{a - b^2 = 0\}$$

$$\Rightarrow \frac{1}{\sqrt{a} - b} = \frac{1}{2} \Rightarrow 2 = \sqrt{a} - b$$

3. **Ans (B)**

$$\frac{dy}{dx} = 3x^2 + 2x - 1$$

slope = 0 (horizontal tangent)

$$3x^2 + 2x - 1 = 0$$

$$x = -1 \text{ or } x = \frac{1}{3}$$

$$\text{when } x = -1, y = 4$$

$$x = 1/3 \text{ or } y = 3 - \frac{5}{27}$$

$$\text{distance} = \frac{32}{27}$$

4. **Ans (B)**

$$I_n = \frac{d^n}{dx^n} (x^n \ln x) = \frac{d^{n-1}}{dx^{n-1}} \left(\frac{d}{dx} (x^n \ln x) \right)$$

$$I_n = \frac{d^{n-1}}{dx^{n-1}} (nx^{n-1} \ln x + x^{n-1})$$

$$= n \frac{d^{n-1}}{dx^{n-1}} \{x^{n-1} \ln x\} + \frac{d^{n-1}}{dx^{n-1}} (x^{n-1})$$

$$I_n = n I_{n-1} + (n-1)!$$

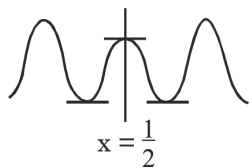
$$I_{100} = 100 I_{99} + 99!$$

PART-3 : MATHEMATICS

SECTION-I (ii)

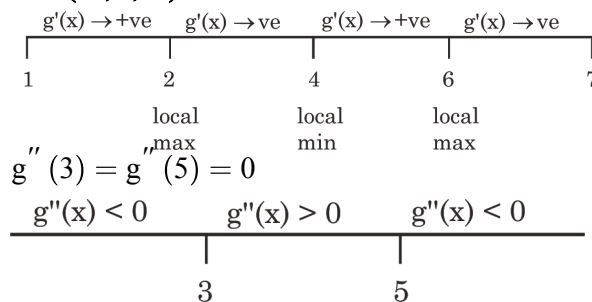
5. **Ans (A,C)**

$$g(x) \text{ is symmetrical about } x = \frac{1}{2}$$



$$\text{i.e. } \lambda = 5$$

7. **Ans (A,C,D)**



8. **Ans (A,B)**

$$x^3 + \frac{1}{x^3} + 3 \cdot x \cdot \frac{1}{x} \left(x + \frac{1}{x} \right) = 3\sqrt{3}$$

$$x^3 + \frac{1}{x^3} = 0 \quad \therefore x^6 = -1$$

$$\text{Now } 1 + x^6 + x^{12} + \dots + x^{6n} = \frac{(x^{6n+6} - 1)}{x^6 - 1}$$

$$= \frac{(-1)^{n+1} - 1}{-2} \begin{cases} 0 & n = \text{odd} \\ 1 & n = \text{even} \end{cases}$$

9. **Ans (B,C)**

$$f(x) = |x-1| + 1 + \frac{1}{|x-1| + 1} - 1$$

$$f(x)1, \{f(x) = 1 \Rightarrow |x-1| + 1 = 1 \Rightarrow x = 1\}$$

$$g(x) = x^3 + 3x + 1$$

$$g'(x) = 3x^2 + 3 \Rightarrow g(x) \text{ is increasing.}$$

$$(A) g(f(x))|_{\min} = g(1) = 5$$

$$(B) f(g(x))|_{\min} = f(x) = 1$$

(C) & (D)

$$\sin^{-1}(f(x)) \text{ is defined at } x = 1 \text{ only.}$$

$$\therefore \tan^{-1}(\tan g(x)) + 4\sin^{-1}(f(x)) = p$$

$$\Rightarrow \tan^{-1}(\tan 5) + 4\sin^{-1}(1) = p$$

$$\Rightarrow 5 - 2\pi + 2\pi = p \Rightarrow p = 5.$$

10. **Ans (B,C)**

$$f(x_0) = \ln \sqrt{2} = \ln(x_0 + \sqrt{1-x_0^2})$$

$$\Rightarrow x_0 = \frac{1}{\sqrt{2}}$$

$$g(x) = \frac{x^2}{f(x)}$$

$$\Rightarrow g'(x) = \frac{f(x) \cdot 2x - x^2 f'(x)}{f^2(x)}$$

$$\Rightarrow g'(x_0) = \frac{2x_0 f(x_0) - x_0^2 f'(x_0)}{f^2(x_0)}$$

$$\{\because f'(x_0) = 0\}$$

$$g'(x_0) = \frac{2x_0}{f(x_0)} = \frac{2 \times \frac{1}{\sqrt{2}}}{\frac{1}{2} \ln 2} = 2\sqrt{2} \log_2 e$$

PART-3 : MATHEMATICS

SECTION-II (i)

1. **Ans (9.00)**

$$\lim_{x \rightarrow 0} \frac{\sqrt{a \sin x + b \cos x + c} - 4}{x} = \frac{1}{8}$$

$$\therefore \sqrt{b+c} - 4 = 0 \Rightarrow b+c = 16 \quad \dots (1)$$

$$\text{Now } \lim_{x \rightarrow 0} \frac{a \sin x + b \cos x + c - 16}{x} \times \frac{1}{\sqrt{a \sin x + b \cos x + c} + 4} = \frac{1}{8}$$

$$\Rightarrow \left(\lim_{x \rightarrow 0} \frac{a \cos x - b \sin x}{1} \right) \times \frac{1}{8} = \frac{1}{8}$$

$$\Rightarrow a = 1$$

As domain of f is $\mathbb{R}$,

$$a \sin x + b \cos x + c \geq 0 \quad \forall x \in \mathbb{R}$$

$$\therefore c - \sqrt{a^2 + b^2} \geq 0$$

$$\Rightarrow c \geq \sqrt{1+b^2} \quad (\text{as } a=1)$$

$$\Rightarrow c^2 \geq 1 + (16-c)^2 \quad (\text{as } b+c=16)$$

$$\Rightarrow 32c \geq 256 + 1$$

$$\Rightarrow c \geq 8 + \frac{1}{32}$$

$\therefore$ Minimum integral value of c is 9.

2. **Ans (8.00)**

$$g(x) = \begin{cases} x^2 + 7x + 9 & x < -2 \\ px + q & x \geq -2 \end{cases}$$

$$\lim_{x \rightarrow -2} g(x) = -1 = -2p + q \quad \dots (1)$$

Also LHD = RHD at $x = -2$

$$\therefore 2(-2) + 7 = p \Rightarrow p = 3$$

$$\therefore q = 5 \Rightarrow \boxed{p+q=8}$$

3. **Ans (2.00)**

Let $x = \tan \theta$, $x \in \mathbb{N} \Rightarrow \theta \in I^{\text{st}}$ quadrant

$$\therefore f(x) = \sin^{-1}(\sin \theta) - \cos^{-1}(\sin \theta) - \frac{\pi}{4}$$

$$= \theta - \left(\frac{\pi}{2} - \theta \right) - \frac{\pi}{4}$$

$$= 2\theta - \frac{3\pi}{4}$$

$$= 2\tan^{-1}x - \frac{3\pi}{4}$$

Now $\text{sgn}(f(x)) = -1 \Rightarrow f(x) < 0$

$$\therefore 2\tan^{-1}x - \frac{3\pi}{4} < 0 \Rightarrow \tan^{-1}x < \frac{3\pi}{8}$$

$$\therefore x < \tan\left(\frac{3\pi}{8}\right) = \sqrt{2} + 1, \Rightarrow x < \sqrt{2} + 1$$

as $x \in \mathbb{N}$, $x = 1$ or 2

4. **Ans (1.00)**

$f(x) = 2\tan^{-1}x - \frac{3\pi}{4}$, $x \in \mathbb{N}$, $f(x)$ is increasing function

$\therefore f(x)$ is minimum at $x = 1$

$$\therefore f_{\min} = 2\tan^{-1}(1) - \frac{3\pi}{4} = -\frac{\pi}{4}$$

6. **Ans (2)**

$$y - f(\alpha) = \frac{f(\beta) - f(\alpha)}{(\beta - \alpha)}(x - \alpha)$$

$$-f(\alpha)(\beta - \alpha) = (f(\beta) - f(\alpha)) \cdot (9\alpha\beta - \alpha)$$

$$-\beta f(\alpha) + \alpha f(\alpha) = 9\alpha\beta f(\beta) - \alpha f(\beta) - 9\alpha\beta f(\alpha) + \alpha f(\alpha)$$

$$-\frac{f(\alpha)}{\alpha} = 9f(\beta) - \frac{f(\beta)}{\beta} - 9f(\alpha)$$

$$9f(\alpha) - \frac{f(\alpha)}{\alpha} = 9f(\beta) - \frac{f(\beta)}{\beta}$$

$$9f(x) - \frac{f(x)}{x} = k$$

$$f(x) = \frac{kx}{9x-1} \quad \therefore f(1) = \frac{5}{8}$$

$$f(x) = \frac{5x}{9x-1} \quad \therefore k = 5$$

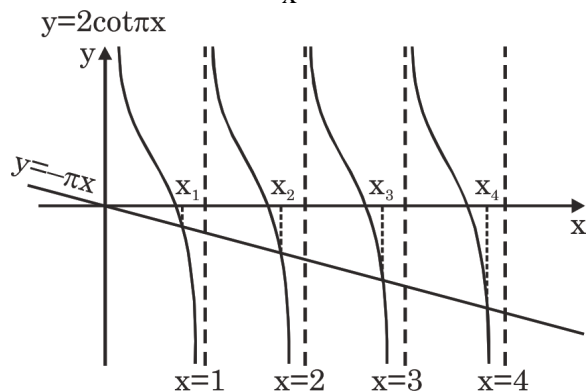
PART-3 : MATHEMATICS

SECTION-II (ii)

7. **Ans (5)**

$$f'(x) = \frac{-x^2 \pi \sin \pi x - (\cos \pi x) 2x}{x^4}$$

$$f'(x) = \frac{-\sin \pi x (2 \cot \pi x - (-\pi x))}{x^3}$$



$$f'(x) = \begin{array}{c} \ominus \oplus \ominus \oplus \ominus \oplus \ominus \oplus \ominus \oplus \ominus \\ x_1 \ x_2 \ x_3 \ x_4 \ x_5 \ x_6 \ x_7 \ x_8 \ x_9 \ x_{10} \end{array}$$

8. **Ans (9)**

$$f(1) = \frac{1}{2}$$

$$\text{Now } g(f(x)) = x$$

Differentiate with respect to x

$$g'(f(x)) \cdot f'(x) = 1$$

$$g'(f(x)) = \frac{1}{f'(x)}$$

Again differentiate w.r.to x

$$g''(f(x)) \cdot f'(x) = -\frac{1}{(f'(x))^2} \cdot f''(x)$$

$$g''(f(x)) = -\frac{1}{(f'(x))^3} \cdot f''(x)$$

Put x = 1

$$g''\left(\frac{1}{2}\right) = \frac{-f''(1)}{(f'(1))^3} \quad \dots (1)$$

$$\text{Now } f(x) = \frac{x^3}{2} + \tan^{-1}(x) - \frac{\pi}{4}$$

$$f'(x) = \frac{3}{2}x^2 + \frac{1}{(1+x^2)} \Rightarrow \boxed{f'(1) = 2}$$

$$f''(x) = 3x - \frac{1}{(1+x^2)^2} (2x)$$

$$f''(1) = 3 - \frac{1}{2} = \frac{5}{2}$$

Now from (1)

$$g''\left(\frac{1}{2}\right) = -\frac{5/2}{8}$$

$$16g''\left(\frac{1}{2}\right) - 4 = -9$$

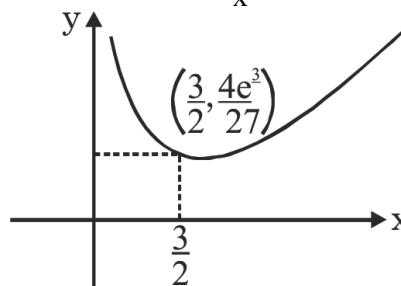
$$\left|16g''\left(\frac{1}{2}\right) - 4\right| = 9$$

9. **Ans (5)**

$$k \leq \frac{e^{2x}}{x^3}$$

$$\text{Now } f(x) = \frac{e^{2x}}{x^3}$$

$$f'(x) = \frac{x^3(2e^{2x}) - e^{2x}(3x^2)}{x^6}$$



$$f'(x) = \frac{e^x[2x-3]}{x^4}$$

$$k \leq \frac{8e^3}{27}$$

$$M = \frac{8e^3}{27}$$

$$[M] = 5$$



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE (Main)
ALLEN COMMON TEST
14-08-2022

JEE(Main + Advanced) : Enthusiast Course [Phase : S, TRAS, I(A), I, II, III & LIVE- I(A), I, I(E)]

ANSWER KEY

PAPER-2

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	D	C	C	C	D	A	A	D	B
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	D	C	B	B	B	C	D	D	B	A
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	9	2	6	13	2	1	4	2	3	8

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	B	C	D	B	D	B	C	D	C
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	A	B	A	C	B	C	D	B	C	B
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	5	6	3	8	5	3	3	4	1	180

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	B	B	C	B	D	C	A	B	D
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	C	B	B	D	B	C	D	D	A	B
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	5	2	9	8	8	0	3	4	3	3

HINT – SHEET

PART-1 : PHYSICS

SECTION-I

1. **Ans (C)**

$$v^2 = 3 - 4x^2$$

$$\Rightarrow v = 2\sqrt{\left(\frac{\sqrt{3}}{2}\right)^2 - x^2} = \omega\sqrt{a^2 - x^2}$$

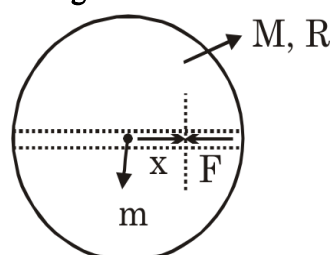
$$\Rightarrow \omega = 2$$

2. **Ans (D)**

$$\text{Use } \vec{E} = - \left[\frac{\partial v}{\partial x} \hat{i} + \frac{\partial v}{\partial y} \hat{j} \right]$$

3. **Ans (C)**

Along diameter



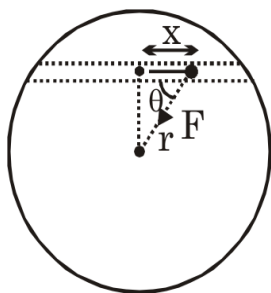
$$F = \left(\frac{GMx}{R^3} \right) m$$

$$= \left(\frac{GMm}{R^3} \right) \cdot x$$

$$T = 2\pi\sqrt{\frac{R^3}{GM}}$$

$$\boxed{T = 2\pi\sqrt{\frac{R^3}{GM}}}$$

Along Chord



F_{net} towards equilibrium = $F \cos \theta$

$$= m \left(\frac{GM}{R^3} \cdot r \right) \cdot \cos \theta = m \left(\frac{GM}{R^3} \right) \cdot x$$

$$T = 2\pi \sqrt{\frac{R^3}{GM}}$$

4. **Ans (C)**

The current in R

$$i_1 = \frac{V}{R} e^{-t/RC}$$

Current in 2R

$$i_2 = \frac{V}{2R} e^{-t/6RC}$$

So if $i_1 = i_2$

$$\Rightarrow \frac{V}{R} e^{-t/RC} = \frac{V}{2R} e^{-t/6RC}$$

$$\Rightarrow e^{-t/RC} = \frac{1}{2} e^{-t/6RC}$$

$$\Rightarrow \frac{-t}{RC} = -\ln 2 - \frac{t}{6RC}$$

$$\Rightarrow t = \frac{6}{5} RC \ln 2$$

5. **Ans (C)**

After closing the switch common potential.

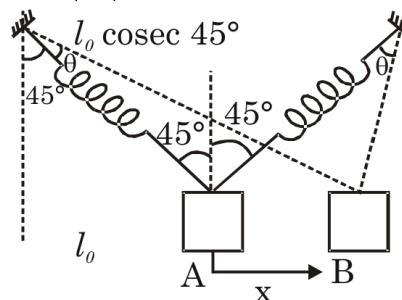
$$V = \frac{4 \times 100 - 2 \times 100}{4 + 2} = \frac{200}{6} = \frac{100}{3}$$

$$U_i = \frac{1}{2} (4 + 2) \times (100)^2 = 3 \times 10^4 \mu\text{J}$$

$$U_f = \frac{1}{2} (4 + 2) \times \left(\frac{100}{3} \right)^2 = \frac{3 \times 10^4}{9} \mu\text{J}$$

$$U_f : U_i = \frac{1}{9} : 1 = 1 : 9$$

6. **Ans (D)**



Force on block at position B along floor

$$F = k[l_0 \csc 45^\circ - l_0 + x \cos 45^\circ] \cos 45^\circ - k[l_0 \csc 45^\circ - l_0 - x \cos 45^\circ] \cos 45^\circ$$

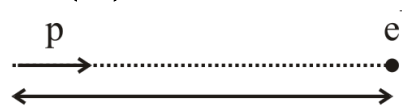
$$m\vec{a} = - (2k \cos^2 45^\circ) \vec{x}$$

$$\vec{a} = - \left(\frac{2k \cos^2 45^\circ}{m} \right) \vec{x}$$

So Time period of SHM of block

$$T = 2\pi \sqrt{\frac{m}{2k \cos^2 45^\circ}} = 2\pi \sqrt{\frac{2}{2 \times 200 \times \left(\frac{1}{\sqrt{2}} \right)^2}} = \frac{\pi}{5} \text{ s}$$

7. **Ans (A)**



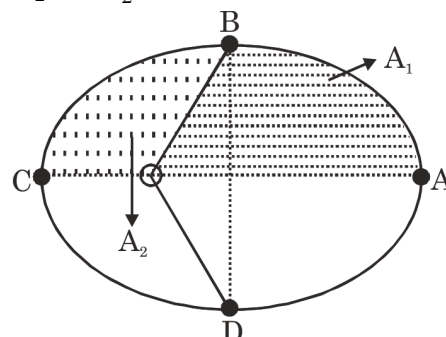
$$K_i + U_i = K_f + U_f$$

$$K_0 + 0 = K - \frac{ep}{4\pi\epsilon_0 r^2}$$

$$K = K_0 + \frac{ep}{4\pi\epsilon_0 r^2}$$

8. **Ans (A)**

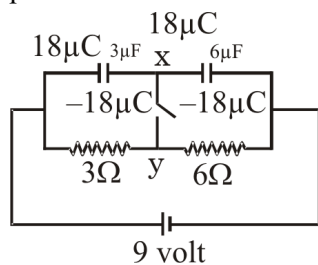
$$\frac{T_1}{T_2} = \frac{A_1}{A_2}$$



$$\begin{aligned} &= \frac{\pi ab}{4} + \frac{1}{2} \cdot ae \cdot b \\ &= \frac{\pi ab}{4} - \frac{1}{2} \cdot ae \cdot b \\ &= \frac{\pi ab + 2aeb}{\pi ab - 2aeb} \\ &= \frac{\pi + 2e}{\pi - 2e} \end{aligned}$$

9. **Ans (D)**

Before closing the switch charge on capacitor plates are shown as



Now after closing the switch in equilibrium state.

i. From battery is given as

$$i = 9/9 = 1 \text{ Amp}$$

So P.D across capacitors will be

$$V_1 = 3 \times 1 = 3V$$

$$V_2 = 6 \times 1 = 6V$$

So charges on $3\mu F$ and $6\mu F$ will be

$$q_1 = 3 \times 3 = 9\mu C$$

$$q_2 = 6 \times 6 = 36\mu C$$

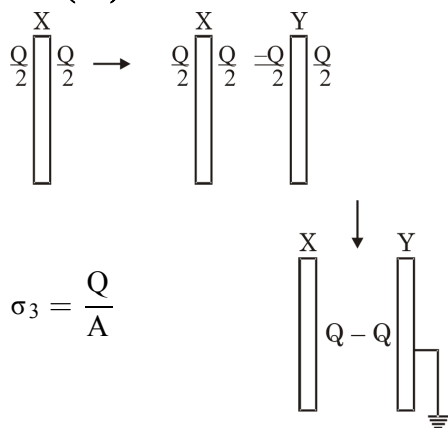
Charge on junction x on the two capacitors plate

$$\Delta q = -9 + 36 = 27\mu C$$

While the initial charge between two plates was $-18 + 18 = 0$

So finally the charge flown from y to x will be $27\mu C$,

10. **Ans (B)**



$$\sigma_3 = \frac{Q}{A}$$

11. **Ans (D)**

$$\text{Average KE w.r.t. time} = \frac{1}{4} KA^2$$

$$\text{Average PE w.r.t. time} = \frac{1}{4} KA^2$$

$$\text{Average KE w.r.t. space} = \frac{1}{3} KA^2$$

$$\text{Average PE w.r.t. space} = \frac{1}{6} KA^2$$

12. **Ans (C)**

$$P \cdot \epsilon_1 = \frac{Q^2}{8\pi\epsilon_0 R} \text{ and } P \cdot \epsilon_2 = \frac{Q^2}{8\pi\epsilon_0 R} - \frac{q^2}{4\pi\epsilon_0 a}$$

$$W = \Delta P \cdot \epsilon = -\frac{q^2}{4\pi\epsilon_0 a}$$

13. **Ans (B)**

$$\mu R \sin \alpha = mvr_{\max}$$

$$\frac{1}{2} \mu u^2 - \frac{GMm}{R} = \frac{1}{2} mv^2 - \frac{GMm}{r_{\max}}$$

$$r_{\max} = R/2 + R \text{ from centre of earth}$$

$$u = \sqrt{\frac{GM}{R}}$$

14. **Ans (B)**

For first case, $V_{AB} = E$

So for the deflection point $E_0 = V_{AC}$

$$\Rightarrow E_0 = \frac{20}{100} \times V_{AB} = \frac{E}{5}$$

For second case -

$$V_{AB} = \left(\frac{10}{10+x} \right) E$$

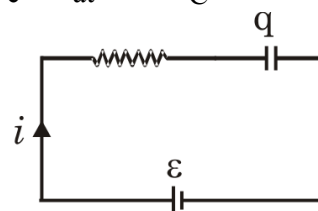
$$E_0 = \frac{30}{100} \times V_{AB} = \frac{30}{100} \times \left(\frac{10E}{10+x} \right)$$

$$\Rightarrow \frac{E}{5} = \frac{3}{10} \left(\frac{10E}{10+x} \right) \Rightarrow \boxed{x = 5}$$

15. **Ans (B)**

$$\epsilon - iR - \frac{q}{c} = 0$$

$$\epsilon = iR + \frac{q}{c} = \frac{dq}{dt} \cdot R + \frac{q}{C}$$



$$\epsilon = R \frac{dq}{dt} + \frac{q}{c} \quad \dots (1)$$

$$K_3 = \frac{1}{K_1} \frac{dq}{dt} + K_2 q \quad \dots (2)$$

On comparing (1) and (2)

$$K_3 = \epsilon, \quad K_1 = \frac{1}{R}, \quad K_2 = \frac{1}{C}$$

$$C = RC = \frac{1}{K_1 K_2}$$

$$Q = C\epsilon = \frac{K_3}{K_1}$$

16. **Ans (C)**

Equations of SHM

$$x_p = a \sin \omega t \quad [\omega = 2\pi f]$$

$$x_Q = a\sqrt{3} + a \sin(\omega t + \phi)$$

$$x_Q - x_P = a\sqrt{3} + 2a \cos\left(\omega t + \frac{\phi}{2}\right) \sin \frac{\phi}{2}$$

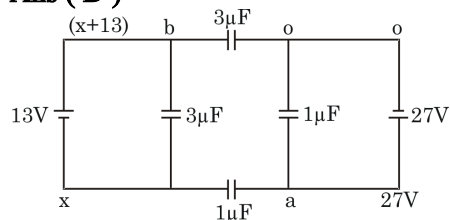
$x_Q - x_P$ is maximum when $\cos\left(\omega t + \frac{\phi}{2}\right)$ is 1 (maximum)

$$2\sqrt{3}a = \sqrt{3}a + 2a \sin \frac{\phi}{2}$$

$$\Rightarrow \sin \frac{\phi}{2} = \frac{\sqrt{3}}{2}$$

$$\Rightarrow \phi = \frac{2\pi}{3} \text{ or } \frac{4\pi}{3}$$

17. **Ans (D)**



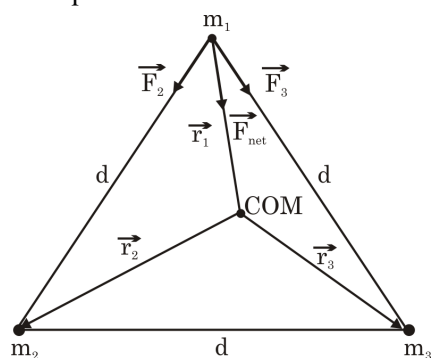
$$(x + 13)3 + (x - 27)1 = 0$$

$$x = -3$$

$$\therefore 17V$$

18. **Ans (D)**

Centre of mass of this three particle system will not be the centroid of the triangle. Because masses are not equal.



$$m_1 \vec{r}_1 + m_2 \vec{r}_2 + m_3 \vec{r}_3 = 0$$

$$-m_1 \vec{r}_1 = m_2 \vec{r}_2 + m_3 \vec{r}_3$$

$$\vec{F}_{\text{net}} = \vec{F}_2 + \vec{F}_3 = -m_1 \omega^2 \vec{r}_1$$

$$\frac{Gm_1 m_2}{d^3} (\vec{r}_2 - \vec{r}_1) - \frac{Gm_1 m_3}{d^3} (\vec{r}_3 - \vec{r}_1) = -m_1 \omega^2 \vec{r}_1$$

$$\frac{Gm_1}{d^3} \{ \underline{m_2 \vec{r}_2} - m_2 \vec{r}_1 - \underline{m_3 \vec{r}_3} + m_3 \vec{r}_1 \} = -m_1 \omega^2 \vec{r}_1$$

$$\frac{G}{d^3} \{ -m_1 \vec{r}_1 - m_2 \vec{r}_1 - m_3 \vec{r}_1 \} = -\omega^2 \vec{r}_1$$

$$\boxed{\sqrt{\frac{G(m_1 + m_2 + m_3)}{d^3}} = \omega}$$

19. **Ans (B)**

$$R_{\text{eq}} = \left(\frac{1}{16} + \frac{1}{8} + \frac{1}{16} \right)^{-1} + 1 = 5\Omega$$

Current supplied by the battery is given as

$$I = \frac{V}{R_{\text{eq}}} = \frac{60}{5} = 12 \text{ Amp}$$

$$\text{Now } \Delta V_{XY} = 60 - 12 \times 1 = 48 \text{ volt}$$

$$\text{So } I_{6\Omega} = \frac{V_{xy}}{16} = \frac{48}{16} = 3 \text{ Amp}$$

20. **Ans (A)**

$$\text{Force on plate} = \frac{Q^2}{2A\epsilon_0}$$

at equilibrium

$$mg = \frac{Q^2}{2A\epsilon_0}$$

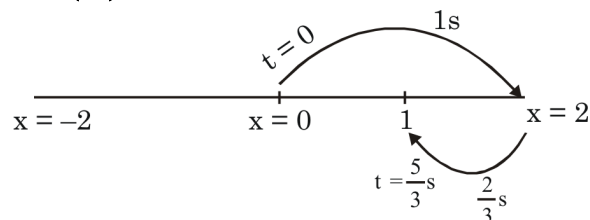
$$Q^2 = 2A\epsilon_0 mg$$

$$Q = \sqrt{2mgA\epsilon_0}$$

PART-1 : PHYSICS

SECTION-II

1. **Ans (9)**

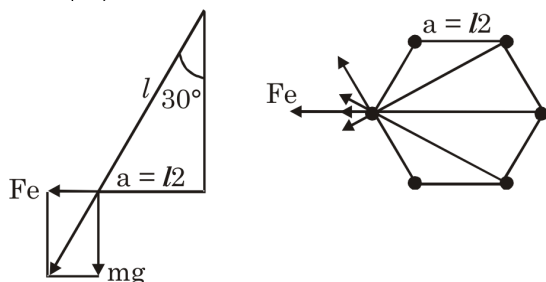


$$x = 2 \sin \frac{\pi}{2} \cdot \frac{5}{3}$$

$$x = 1$$

$$\text{Now average speed} = \frac{2+1}{\frac{5}{3}} = \frac{9}{5} \text{ s}$$

2. Ans (2)



Electric force on a bob is given by

$$F_e = \frac{Q^2}{4\pi\epsilon_0 a^2} \left[2 \times \cos(60^\circ) + 2 \times \frac{1}{3} \times \cos(30^\circ) + \frac{1}{4} \right]$$

$$F_e = \frac{Q^2}{4\pi\epsilon_0 a^2} \left[1 + \frac{1}{\sqrt{3}} + \frac{1}{4} \right] \quad \dots (A)$$

At equilibrium

$$\tan(30^\circ) = \frac{F_e}{mg}$$

$$F_e = \frac{1}{\sqrt{3}} mg \quad \dots (B)$$

From A and B

$$\frac{1}{\sqrt{3}} mg = \frac{Q^2}{4\pi\epsilon_0 a^2} \left[1 + \frac{1}{\sqrt{3}} + \frac{1}{4} \right]$$

$$\frac{4\pi\epsilon_0 m g a^2}{Q^2} = \left[\sqrt{3} + 1 + \frac{\sqrt{3}}{4} \right]$$

$$\frac{\pi\epsilon_0 m g l^2}{Q^2} = \left[1 + \frac{5\sqrt{3}}{4} \right]$$

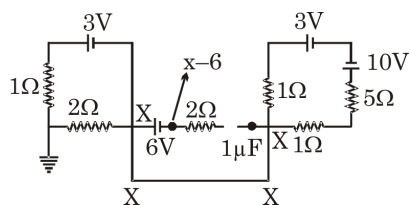
$$\frac{\pi\epsilon_0 m g l^2}{Q^2} = \left[x + \frac{y\sqrt{z}}{4} \right]$$

$$\frac{x + y + z}{4} = 2.25$$

3. Ans (6)

At steady state there is no current in middle loop

because of capacitor behave as open circuit.



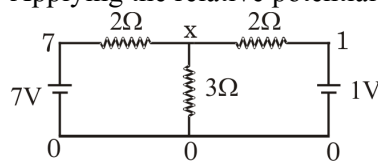
Potential difference across capacitor = $(x - 6) - x = 6V$

Charge = $Q = CV = 6 \times 1\mu F$

$Q = 6\mu C$

4. Ans (13)

Applying the relative potential analysis



$$\text{So } \frac{x-7}{2} + \frac{x}{3} + \frac{x-1}{2} = 0 \Rightarrow x = 3 \text{ volt}$$

So current from 7V battery = 2A

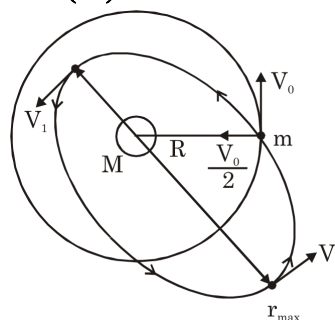
So $P_{7V} = 7 \times 2 = 14 \text{ watt}$

and current by 1V battery = 1 Amp

So $P_{1V} = -1 \times 1 = -1 \text{ watt}$

So $P_1 + P_2 = 14 - 1 = 13 \text{ watt}$

5. Ans (2)



Angular momentum conservation

$$mV_0R = mVr \quad \dots (1)$$

Energy Conservation

$$-\frac{GMm}{R} + \frac{1}{2}m \left(V_0^2 + \frac{V_0^2}{4} \right) = -\frac{GMm}{r} + \frac{1}{2}mV^2 \quad \dots (2)$$

$$-\frac{GMm}{R} + \frac{1}{2}m \left(\frac{5}{4}V_0^2 \right) = -\frac{GMm}{r} + \frac{1}{2}mV^2$$

$$-\frac{GMm}{R} + \frac{1}{2}m \cdot \frac{5}{4} \cdot \left(\frac{GM}{R} \right) = -\frac{GMm}{r} + \frac{1}{2}m \left(\frac{V_0^2 R^2}{r^2} \right)$$

$$-\frac{3}{8} \cdot \frac{GMm}{R} = -\frac{GMm}{r} + \frac{1}{2}m \cdot \left(\frac{GM}{R} \cdot \frac{R^2}{r^2} \right)$$

$$-3r^2 = -8Rr + 4R^2$$

$$3r^2 - 8Rr + 4R^2 = 0$$

$$r_1 + r_2 = \frac{8R}{3}$$

$$r_1 r_2 = \frac{4R^2}{3}$$

$$(r_2 - r_1)^2 = (r_1 + r_2)^2 - 4r_1 r_2 = \left(\frac{8R}{3} \right)^2 - \frac{16R^2}{3}$$

$$= \frac{64R^2}{9} - \frac{16R^2}{3} = \frac{16R^2}{9}$$

$$r_2 - r_1 = \frac{4R}{3}$$

$$r_2 + r_1 = \frac{8R}{3}$$

$$r_2 = 2R; r_1 = \frac{2R}{3}$$

6. Ans (1)

$$\epsilon = \frac{KQ^2}{d}$$

$$\epsilon' = \frac{KQ^2}{d} - \frac{2KQ^2}{d} = \frac{-KQ^2}{d}$$

$$\epsilon' = -\epsilon$$

7. Ans (4)

$$12 - \frac{Q}{3} - \frac{Q}{6} = 0$$

$$Q = 24\text{mC}$$

$$V_{AP} = \frac{24}{3} = 8\text{V}$$

$$V_P = 12 - 8 = 4\text{V}$$

8. Ans (2)

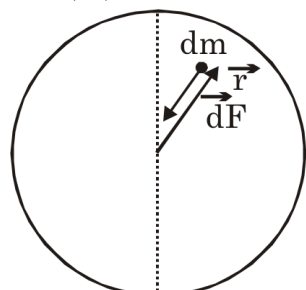
For the electric field, $V = \frac{E_0 x^2}{2\ell} - \frac{E_0 y^2}{2\ell} + K$

from energy conservation

$$-\frac{QE_0}{\ell} \frac{\ell^2}{2} + k + \frac{1}{2}Mv^2 = \frac{QE_0}{\ell} \frac{\ell^2}{8} \times 2 + K$$

$$v = 2 \text{ m/s}$$

9. Ans (3)



$$\vec{dF} = (dm) \vec{g}$$

$$\int \vec{dF} = \int dm \cdot \frac{GM}{R^3} \cdot (-\vec{r})$$

$$= -\frac{Gm}{R^3} \int \vec{r} \cdot dm$$

$$\frac{\int \vec{r} dm}{\int dm} = \vec{r}_{\text{com}}$$

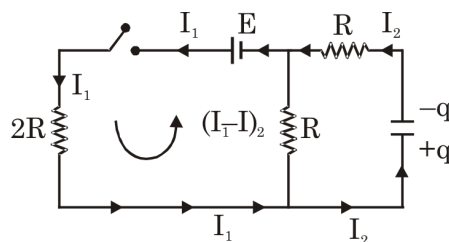
$$= -\frac{GM}{R^3} \cdot \vec{r}_{\text{com}} \cdot \int dm \quad \left(\int dm = \frac{M}{2} \right)$$

$$\vec{F} = -\frac{GM^2}{2R^3} \cdot \vec{r}_{\text{com}}$$

$$r_{\text{com}} = 3R/8$$

$$F = \frac{3GM^2}{16R^2}$$

10. Ans (8)



From KVL $+E - I_1(2R) - (I_1 - I_2)R = 0$

$$3I_1 - I_2 = E/R \quad \dots(1)$$

and $I_2R - (I_2 - I_1)R - q/C = 0$

$$I_1 - 2I_2 = q/RC \quad \dots(2)$$

From (1) and (2)

$$I_2 = \frac{CE - 3q}{5RC} = \frac{dq}{dt}$$

$$\text{So } \int_0^q \frac{dq}{CE - 3q} = \int_0^t \frac{dt}{5RC}$$

$$\Rightarrow q = \frac{CE}{3} (1 - e^{-3t/5RC})$$

Hence $a = 3$ and $b = 5$

So $a + b = 3 + 5 = 8$

PART-2 : CHEMISTRY

SECTION-I

1. Ans (B)

Moles of CO_2 formed $= 0.5 \times 0.2 \times 0.2$

$= 0.02$ moles

Mass of $\text{CO}_2 = 0.02 \times 44$

$= 0.88$ gram

2. Ans (B)

$$\Delta T_f = k_f \times m \times i$$

$$5.58 = 1.86 \times 1 \times i$$

$i = 3$, so, three ions form

3. Ans (C)

$$\Delta T_f = k_f \times m$$

$$15 = 1.86 \times \frac{x}{62} \times \frac{1000}{500}$$

$$x = \frac{15 \times 62 \times 500}{1.86 \times 1000} = 250$$

4. Ans (D)

Let W gas of SO₂ and O₂ are taken

$$\text{moles of SO}_2 = \frac{W}{64}; \text{ moles of O}_2 = \frac{W}{32}$$

$$\text{molecules of O}_2 = \frac{W N_A}{32}; \text{ molecules of SO}_2 = \frac{W N_A}{64}$$

hence molecules of O₂ > molecules of SO₂

since moles of O₂ > moles of SO₂, hence

volume of O₂ at STP > volume of SO₂ at STP.

5. Ans (B)

First case :

$$K = \frac{1}{t_{\frac{1}{3}}} \ln \frac{a_0}{\frac{2a_0}{3}}$$

$$K = \frac{1}{t_{\frac{1}{3}}} \ln \frac{3}{2} \dots\dots(1)$$

In second case

$$K = \frac{1}{t_{\frac{2}{3}}} \ln \frac{a_0}{\frac{a_0}{3}}$$

$$K = \frac{1}{t_{\frac{2}{3}}} \ln 3 \dots\dots(2)$$

from (1) and (2)

$$t_{\frac{2}{3}} - t_{\frac{1}{3}} = \frac{\ln 3}{k} - \frac{\ln \frac{3}{2}}{k}$$

$$200 = \frac{\ln 2}{k}$$

$$k = \frac{\ln 2}{200}$$

$$\text{Hence, } k = \frac{1}{t} \ln \frac{a_0}{\frac{a_0}{4}}$$

$$\frac{\ln 2}{200} = \frac{1}{t} \ln 4$$

$$\frac{\ln 2}{200} = \frac{2 \ln 2}{t}$$

$$\therefore t = 400$$

6. Ans (D)

$$x_{\text{NaOH}} = 0.2$$

$$x_{\text{H}_2\text{O}} = 0.8$$

$$\text{i.e. } n_{\text{NaOH}} = 0.2$$

$$n_{\text{water}} = 0.8$$

$$n_{\text{NaOH}} = 1, n_{\text{H}_2\text{O}} = 4$$

$$W_{\text{NaOH}} = 1 \times 40 = 40$$

$$W_{\text{H}_2\text{O}} = 4 \times 18 = 72$$

$$W_{\text{soln}} = 40 + 72 = 112 \text{ gram}$$

$$v = \frac{M}{d} = \frac{112}{14} = 80 \text{ ml}$$

$$M = \frac{n_{\text{NaOH}}}{V} = \frac{1}{80 \times 10^{-3}} = \frac{1000}{80} = 12.5 \text{ M}$$

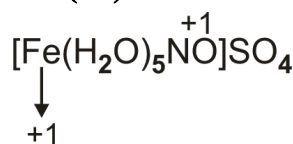
11. Ans (A)

Fe(CO)₅ and [Mn(CO)₅]⁺ due to synergic bonding CO bond order decreases.

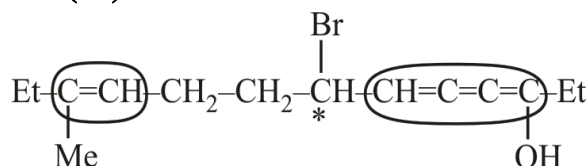
12. Ans (B)

It has square planar geometry and hydrogen bonding between two DMG.

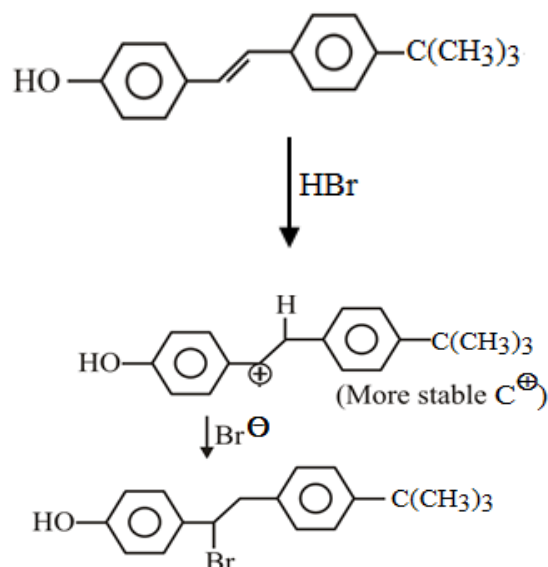
13. Ans (A)



14. Ans (C)



15. Ans (B)



16. **Ans (C)**

Reaction (A) occur via $S_N2 \Rightarrow$ Inversion takes place.

Reaction (B) occur via S_N1 mechanism $\Rightarrow$ so, rearrangement of carbocation give the correct product.

Reaction (C) occur via S_Ni mechanism $\Rightarrow$ Retention takes place.

Reaction (D) occur via S_N2 mechanism $\Rightarrow$ Inversion take place.

17. **Ans (D)**

In polar protic solvent $\Rightarrow S_N1$ mechanism favoured.

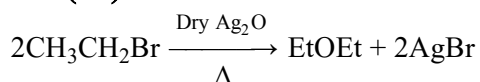
In polar aprotic solvent $\Rightarrow S_N2$ mechanism favoured.

Due to hydration (solvation) in polar protic solvent, the nucleophilicity of smaller ion decreases.

Reaction can pass through S_N2 mechanism upto some extent in case of S_N1 mechanism. So 100% racemisation is not possible always in S_N1 mechanism.

Down the group, basic strength $\downarrow$ es is polar protic as well as polar aprotic solvent.

20. **Ans (B)**



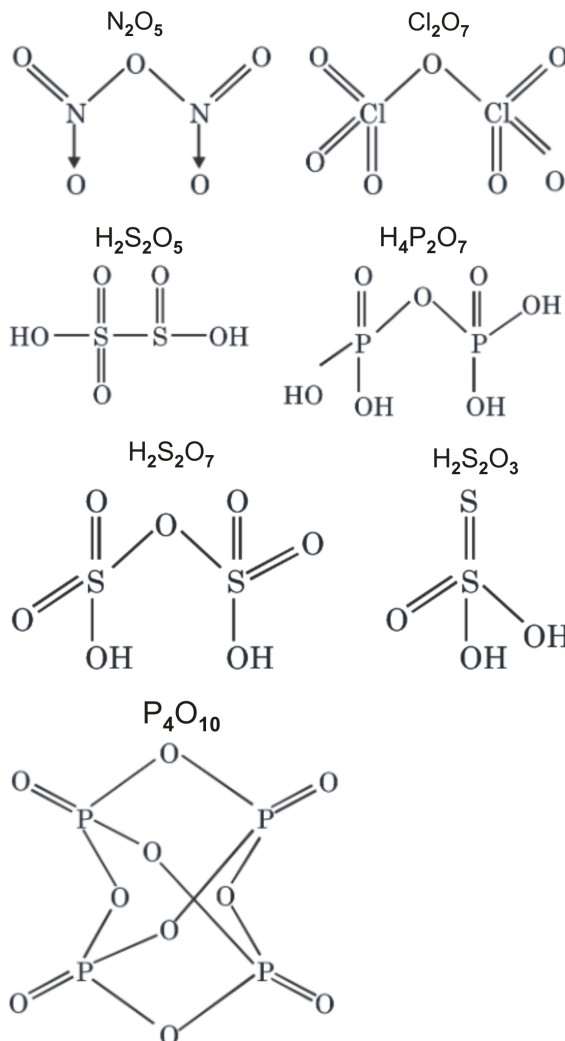
PART-2 : CHEMISTRY

SECTION-II

1. **Ans (5)**

1, 3, 4, 5, 7

2. **Ans (6)**



3. **Ans (3)**

3, 5, 6

4. **Ans (8)**

Co-ordination number 6

Oxidation number 2

5. **Ans (5)**

C-C-C-C-C-OH ... (1)

C-C-C-C-OH ... (1)

C-C-C-C-OH ... (2)

C-C-C-OH ... (1)

6. **Ans (3)**

3, 4, 6

8. **Ans (4)**

$$\frac{EO_x}{E} = \frac{4}{16x} = \frac{1}{4x}$$

$$\frac{EO_y}{E} = \frac{3}{16y} = \frac{3}{16y}$$

$$\frac{y}{x} = \frac{6}{2} = \frac{3}{1}$$

$$x = 1, y = 3, x + y = 4$$

10. **Ans (180)**

$$A \rightarrow 2B + C$$

$$90 - -$$

$$900 - x \quad 2x \quad x$$

$$P_{\text{Total}} = 90 + 2x$$

$$K = \frac{1}{t} \ln \frac{90}{90 - x}$$

$$1.15 \times 10^{-3} = \frac{1}{600} \ln \frac{90}{90 - x}$$

$$0.690 = \ln \frac{90}{90 - x}$$

$$\frac{90}{90 - x} = 2$$

$$x = 45$$

$$P_{\text{Total}} = 90 + 90 = 180$$

PART-3 : MATHEMATICS

SECTION-I

1. **Ans (B)**

$$\lim_{x \rightarrow 2^{-1/4}} \frac{\cos \cos^{-1} \left(\sqrt{1 - x^4} \right) - x^2}{1 - \tan \tan^{-1} \left(\frac{\sqrt{1 - x^4}}{x^2} \right)}$$

$$= \lim_{x \rightarrow 2^{-1/4}} \frac{\sqrt{1 - x^4} - x^2}{1 - \frac{\sqrt{1 - x^4}}{x^2}}$$

$$= \lim_{x \rightarrow 2^{-1/4}} (-x^2)$$

$$= -(2^{-1/4})^2 = -\frac{1}{\sqrt{2}}$$

2. **Ans (B)**

$$f(x) = [x] + [2x] + [3x] + \dots + [nx] - (x + 2x + 3x + \dots + nx)$$

$$= -(\{x\} + \{2x\} + \{3x\} + \dots + \{nx\})$$

$$\text{The period of } f(x) = \text{LCM} \left(1, \frac{1}{2}, \frac{1}{3}, \dots, \frac{1}{n} \right) = 1$$

3. **Ans (B)**

$$8f(x) + 6f\left(\frac{1}{x}\right) = x + 5 \quad \dots(1)$$

Replace x with $\frac{1}{x}$

$$8f\left(\frac{1}{x}\right) + 6f(x) = \frac{1}{x} + 5 \quad \dots(2)$$

from (1) and (2)

$$f(x) = \frac{8x}{28} - \frac{6}{28x} + \frac{10}{28}$$

$$\text{Now, } y = \frac{8x^4}{28} - \frac{6x^2}{28} + \frac{10x^3}{28}$$

$$\frac{dy}{dx} = \frac{32x^3}{28} - \frac{12x}{28} + \frac{30x^2}{28}$$

$$\frac{dy}{dx} \Big|_{x=-1} = \frac{-32}{28} + \frac{12}{28} + \frac{30}{28} = \frac{10}{28} = \frac{5}{14}$$

4. **Ans (C)**

$$\frac{dy}{dx} = f' \left(\frac{5x+3}{3x+5} \right) \cdot \left(\frac{(3x+5)(5) - (5x+3)3}{(3x+5)^2} \right)$$

$$= \cot \left(\frac{5x+3}{3x+5} \right)^2 \left(\frac{16}{(3x+5)^2} \right)$$

$$= 16 \cot \left(\frac{5x+3}{3x+5} \right)^2 \cdot \frac{1}{(3x+5)^2}$$

5. **Ans (B)**

$$\text{We have } f(x-y) = f(x)f(y) - f(a-x)f(a+y)$$

Putting $x = a$ and $y = x - a$, we get

$$f(a - (x - a)) = f(a)f(x - a) - f(0)f(x) \quad \dots(1)$$

Putting $x = 0$, $y = 0$, we get

$$f(0) = f(0)f(0) - f(a)f(a)$$

$$\Rightarrow f(0) = (f(0))^2 - f(a)^2$$

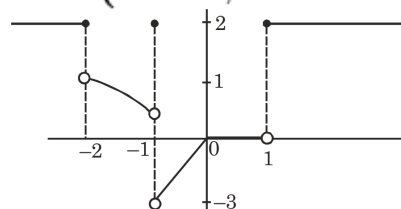
$$\Rightarrow 1 = (1)^2 - f(a)^2$$

$$\Rightarrow f(a) = 0$$

$$\Rightarrow f(2a - x) = -f(x)$$

6. **Ans (D)**

$$f(x) = \begin{cases} \cos(x+2); & x \in (-2, -1) \\ \min\{3x, 0\}; & -1 < x < 1 \\ 2; & \text{otherwise} \end{cases}$$



Discontinuous at $x = -2, -1, 1$

No differentiable at $x = -2, -1, 0, 1$

7. **Ans (C)**

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\cos^2 x - 3 \cos x + 2}{\tan^2 x (3 + 3)} \\ = \lim_{x \rightarrow 0} \frac{(\cos x - 1)(\cos x - 2) \cdot \cos^2 x}{6 \sin^2 x} \\ = \lim_{x \rightarrow 0} \frac{-(\cos x - 2) \cos^2 x}{6(1 + \cos x)} = \frac{1}{6 \times 2} = \frac{1}{12} \end{aligned}$$

8. **Ans (A)**

Let $f(x) = x^3 + ax^2 + bx + c$
 $f'(x) = 3x^2 + 2ax + b$
 $f''(x) = 6x + 2a$
 Now $f(x) + f'(x) + f''(x) = x^3 + 1$
 $\Rightarrow x^3 + (a+3)x^2 + (2a+b+6)x + 2a+b+c = x^3 + 1$

Comparing coefficients :

$$a = -3, \quad b = 0, \quad c = 4$$

$$\therefore f(x) = x^3 - 3x^2 + 4$$

$$\begin{aligned} \therefore \lim_{x \rightarrow 2} \frac{f(x)}{x-2} \\ = \lim_{x \rightarrow 2} \frac{x^3 - 3x^2 + 4}{x-2} \\ = \lim_{x \rightarrow 2} \frac{(x-2)(x^2 - x - 2)}{x-2} = 0 \end{aligned}$$

9. **Ans (B)**

$$\begin{aligned} \cos^{-1}(2x^2 - 1) &= 2\pi - 2\cos^{-1}x \quad (\text{as } x < 0) \\ \Rightarrow \cos^{-1}(2x^2 - 1) - 2\sin^{-1}x &= 2\pi - 2\cos^{-1}x - 2\sin^{-1}x \\ &= 2\pi - 2(\cos^{-1}x + \sin^{-1}x) \\ &= 2\pi - 2 \cdot \frac{\pi}{2} = \pi \end{aligned}$$

10. **Ans (D)**

$$\begin{aligned} a \sin^{-1}x - b \cos^{-1}x &= c \\ \text{We have } b \sin^{-1}x + b \cos^{-1}x &= \frac{b\pi}{2} \\ \Rightarrow (a+b) \sin^{-1}x &= \frac{b\pi}{2} + c \\ \Rightarrow \sin^{-1}x &= \frac{\frac{b\pi}{2} + c}{a+b} \\ \Rightarrow \cos^{-1}x &= \frac{\pi ab + c(a-b)}{a+b} \end{aligned}$$

11. **Ans (C)**

$$\begin{aligned} y' &= \frac{2e^{\sqrt{2x}}}{2\sqrt{2x}} - \frac{2e^{-\sqrt{2x}}}{2\sqrt{2x}} \\ \sqrt{2x} y' &= e^{\sqrt{2x}} - e^{-\sqrt{2x}} \\ \sqrt{2x} y'' + \frac{2}{2\sqrt{2x}} y' &= \frac{e^{\sqrt{2x}}}{\sqrt{2x}} + \frac{e^{-\sqrt{2x}}}{\sqrt{2x}} \\ \sqrt{2x} y'' + \frac{y'}{\sqrt{2x}} &= \frac{y}{\sqrt{2x}} \\ 2xy'' + y' &= y \end{aligned}$$

12. **Ans (B)**

Given $g(x) = f^{-1}(x)$
 So $g(f(x)) = x \Rightarrow g'(f(x)) = \frac{1}{f'(x)}$
 $\Rightarrow g''(f(x))f'(x) = \frac{-f''(x)}{(f'(x))^2}$
 $\Rightarrow g''(f(x)) = \frac{-f''(x)}{(f'(x))^3}$
 at $x = 1 \quad g''(f(1)) = \frac{-(-8)}{2^3} = 1$

13. **Ans (B)**

$$\begin{aligned} f(x) &= \frac{1}{2\{-x\}} - \{1 - \{-x\}\} \\ \{-x\} + \frac{1}{2\{-x\}} &- 1 \\ \geq 2 \sqrt{\frac{1}{2}} - 1 &\geq \sqrt{2} - 1 \\ f_{\min} &= \sqrt{2} - 1 \text{ when } \{-x\} = \frac{1}{\sqrt{2}} \\ -x - n &= \frac{1}{\sqrt{2}} \quad -x = n + \frac{1}{\sqrt{2}} \\ x &= -n - \frac{1}{\sqrt{2}} \end{aligned}$$

14. Ans (D)

$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots$$

$$e^{-x} = 1 - \frac{x}{1!} + \frac{x^2}{2!} - \frac{x^3}{3!} + \frac{x^4}{4!} - \dots$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$$

$$\text{So } e^x + e^{-x} + 2 \cos x$$

$$= 2 \left(1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \frac{x^6}{6!} + \dots + 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots \right)$$

$$= 2 \left(2 + \frac{2x^4}{4!} + \frac{2x^8}{8!} + \frac{2x^{12}}{12!} + \dots \right)$$

$$e^x + e^{-x} + 2 \cos x = 4 \left(1 + \frac{x^4}{4!} + \frac{x^8}{8!} + \frac{x^{12}}{12!} + \dots \right)$$

$$= 4 + \frac{x^4}{6} + 4 \left(\frac{x^8}{8!} + \frac{x^{12}}{12!} + \dots \right)$$

$$e^x + e^{-x} + 2 \cos x - \frac{x^4}{6} - 4 = 4 \left(\frac{x^8}{8!} + \frac{x^{12}}{12!} + \dots \right)$$

$$f(x) = 4 \left(\frac{x^8}{8!} + \frac{x^{12}}{12!} + \frac{x^{16}}{16!} + \dots \right)$$

$$f'(x) = 4 \left(\frac{x^7}{7!} + \frac{x^{11}}{11!} + \frac{x^{15}}{15!} + \dots \right)$$

$$f''(x) = 4 \left(\frac{x^6}{6!} + \frac{x^{10}}{10!} + \frac{x^{14}}{14!} + \dots \right)$$

$$\frac{d^8 x}{dx^8} = 4 \left(x^0 + \frac{x^4}{4!} + \frac{x^8}{8!} + \dots \right)$$

$$\text{at } x = 0 \quad \frac{d^8 x}{dx^8} = 4 \quad \text{that is a non-zero constant}$$

$$\text{So } n = 8$$

15. Ans (B)

Let $(\sqrt{6} \cos \theta, \sqrt{3} \sin \theta)$ is point.

Then $\frac{x}{\sqrt{6}} \cos \theta + \frac{y}{\sqrt{3}} \sin \theta = 1$ is parallel to

$$x + y = 7$$

$$\Rightarrow \frac{\cos \theta}{\sqrt{2}} = \frac{\sin \theta}{1} = \frac{1}{\sqrt{3}}$$

$$\text{Hence, } (a, b) = (2, 1)$$

16. Ans (C)

Indeterminate form of type $0 \times \infty$

Rationalising

$$\lim_{x \rightarrow \frac{\pi}{2}} \tan^2 x \frac{((2 \sin^2 x - 3 \sin x + 4) - (\sin^2 x + \sin x + 1))}{\sqrt{2 \sin^2 x - 3 \sin x + 4} + \sqrt{\sin^2 x + \sin x + 1}}$$

$$\lim_{x \rightarrow \frac{\pi}{2}} \tan^2 x \frac{(\sin^2 x - 4 \sin x + 3)}{\sqrt{3} + \sqrt{3}}$$

$$= \frac{1}{2\sqrt{3}} \lim_{x \rightarrow \frac{\pi}{2}} \sin^2 x \frac{(\sin x - 1)(\sin x - 3)}{(1 - \sin x)(1 + \sin x)}$$

$$= \frac{1}{2\sqrt{3}}$$

17. Ans (D)

Apply $C_1 \rightarrow C_1 - C_3$ and $C_2 \rightarrow C_2 - C_3$

$$f'(x) = \begin{vmatrix} -(b+1) & -(b+2) & 2ax+b+1 \\ b+1 & b+2 & -1 \\ b & b+1 & 2ax+b \end{vmatrix}$$

$R_1 \rightarrow R_1 + R_2$ and $R_3 \rightarrow R_3 - R_2$

$$f'(x) = \begin{vmatrix} 0 & 0 & 2ax+b \\ b+1 & b+2 & -1 \\ -1 & -1 & 2ax+b+1 \end{vmatrix}$$

$$\Rightarrow f'(x) = 2ax + b$$

$$\therefore f(x) = ax^2 + bx + c$$

$$f(0) = 2 \Rightarrow c = 2$$

$$f(1) = 1 \Rightarrow a + b + 2 = 1 \Rightarrow a + b = -1$$

$$f'\left(\frac{5}{2}\right) = 0 \Rightarrow 5a + b = 0, a = \frac{1}{4}, b = -\frac{5}{4}$$

$$f(x) = \frac{1}{4}x^2 - \frac{5}{4}x + 2$$

$$\text{Range is } f(x) \in \left[\frac{7}{16}, \infty\right)$$

18. Ans (D)

Let $f(x) = mx + b$

$$\therefore f(1) \leq f(2) \Rightarrow m \geq 0, \text{ similarly}$$

$$f(3) \geq f(4) \Rightarrow m \leq 0 \Rightarrow m = 0$$

$$\therefore f(0) = f(5) = 5$$

19. Ans (A)

$$2^{1+(\log_2 x)^2} + (x^{\log_2 x})^2 - 3 = 0$$

$$2 \cdot (2^{\log_2 x})^{\log_2 x} + (x^{\log_2 x})^2 - 3 = 0$$

$$2 \cdot x^{\log_2 x} + (x^{\log_2 x})^2 - 3 = 0$$

$$\text{Now put } x^{\log_2 x} = t$$

$$t^2 + 2t - 3 = 0 \Rightarrow (t+3)(t-1) = 0$$

$$\Rightarrow t = 1$$

$$\Rightarrow x^{\log_2 x} = 1$$

$$(\log_2 x)(\log_2 x) = 0 \Rightarrow x_0 = 1$$

$$\therefore \sin^{-1}(x_0) + \tan^{-1}\left(\frac{2x_0}{2 - (x_0)^2}\right) + \cot^{-1}(2)$$

$$= \frac{\pi}{2} + \tan^{-1}2 + \cot^{-1}2 = \pi$$

20. Ans (B)

$$\operatorname{sgn}\left(\left[\frac{15}{1+x^2}\right]\right) = 1$$

$$1 + x^2 \leq 15 \Rightarrow x^2 \leq 14$$

$\therefore$ Number of integral values of x are 7.

PART-3 : MATHEMATICS

SECTION-II

1. Ans (5)

$$\text{LHL : } \lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} (1+ax)^{1/x} \quad (1^\infty \text{ form})$$

$$= \exp\left(\lim_{x \rightarrow 0^-} \left(\frac{1+ax-1}{x}\right)\right) = e^a$$

$$\text{RHL : } \lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} \frac{(x+c)^{1/3} - 1}{(x+1)^{1/2} - 1}$$

$$\text{for finite limit } c = 1 \left(\frac{0}{0} \text{ form}\right)$$

$$\lim_{x \rightarrow 0^+} \frac{(x+1)^{1/3} - 1}{(x+1)^{1/2} - 1} \times \frac{(x+1)^{1/2} + 1}{(x+1)^{1/2} + 1} \cdot \frac{((x+1)^{2/3} + (x+1)^{1/3} + 1)}{(x+1)^{2/3} + (x+1)^{1/3} + 1}$$

$$\lim_{x \rightarrow 0^+} \frac{(x+1-1)(x+1)^{1/2} + 1}{(x+1-1)((x+1)^{2/3} + (x+1)^{1/3} + 1)} = \frac{2}{3}$$

$$\Rightarrow \text{LHL} = \text{RHL} = f(0)$$

$$\Rightarrow e^a = b = \frac{2}{3} \Rightarrow a = \ln\left(\frac{2}{3}\right)$$

$$p + q = 5$$

2. Ans (2)

$$\lim_{n \rightarrow \infty} \frac{5 \cdot 5^n + 3^n - 9^n \cdot 3}{5^n + 2^n + 27 \cdot 9^n} = L$$

$$\lim_{n \rightarrow \infty} \frac{5\left(\frac{5}{9}\right)^n + \left(\frac{3}{9}\right)^n - 3}{\left(\frac{5}{9}\right)^n + \left(\frac{2}{9}\right)^n + 27} = L$$

$$L = \frac{-1}{9} \Rightarrow \boxed{-18L = 2}$$

3. Ans (9)

$$\lim_{x \rightarrow 0} \frac{\sin 3x + A \sin 2x + B \sin x}{x^5} \quad \left(\frac{0}{0}\right) \text{ form}$$

$$(\text{By L'Hospital}) : \lim_{x \rightarrow 0} \frac{3 \cos 3x + 2A \cos 2x + B \cos x}{5x^4},$$

for finite limit

$$3 + 2A + B = 0 \Rightarrow 2A + B = -3 \quad \dots(i)$$

again L'Hospital

$$\lim_{x \rightarrow 0} \frac{-9 \sin 3x - 4A \sin 2x - B \sin x}{20x^3} \quad \left(\frac{0}{0}\right) \text{ form}$$

$$\lim_{x \rightarrow 0} \frac{-27 \cos 3x - 8A \cos 2x - B \cos x}{60x^2}$$

for limit to exist

$$-27 - 8A - B = 0$$

$$8A + B = -27 \quad \dots(ii)$$

From eq. (i) & eq. (ii)

$$2A + B = -3$$

$$8A + B = -27$$

$$-6A = 24$$

$$A = -4 \quad B = 5$$

$$|A| + |B| = 9$$

4. Ans (8)

$$y = f(x^2)$$

$$\frac{dy}{dx} = f'(x^2) \cdot 2x$$

$$\frac{dy}{dx} = 2x \sqrt{2x^4 + 1}$$

$$\text{at } x = \sqrt{2}$$

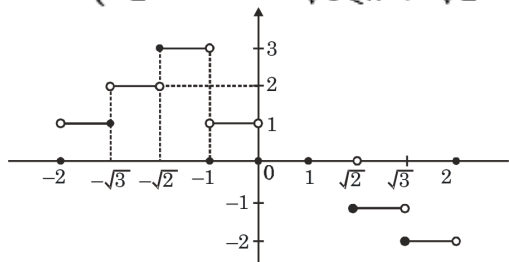
$$\frac{dy}{dx} = \sqrt{2 \cdot 4 + 1} \cdot (2\sqrt{2})$$

$$= 6\sqrt{2}$$

$$a = 6 \quad b = 2 \Rightarrow a + b = 8$$

5. Ans (8)

$$f(x) = \begin{cases} 0 & 0 \leq x < \sqrt{2} \text{ and } x = 2, -1, -2 \\ -1 & \sqrt{2} \leq x < \sqrt{3} \\ -2 & \sqrt{3} \leq x < 2 \\ 1 & -1 < x < 0 \text{ and } -2 < x \leq -\sqrt{3} \\ 3 & -\sqrt{2} \leq x < -1 \\ 2 & -\sqrt{3} \leq x < -\sqrt{2} \end{cases}$$



Total points 8 where $f(x)$ is discontinuous

6. Ans (0)

$$\tan^{-1} \left(\frac{1}{2K^2} \right) = \tan^{-1} \left(\frac{K}{K+1} \right) - \tan^{-1} \left(\frac{K-1}{K} \right)$$

$$\Rightarrow \lim_{n \rightarrow \infty} \sum_{K=1}^{n+1} \tan^{-1} \left(\frac{1}{2K^2} \right) = \lim_{n \rightarrow \infty} \tan^{-1} \left(\frac{n}{n+1} \right) = \frac{\pi}{4}$$

7. Ans (3)

Let $(-1, \alpha)$ be the point on $y = g(x)$

$\Rightarrow (\alpha, -1)$ lies on $y = f(x)$

Also $x = f(g(x))$

$$\Rightarrow g'(-1) = \frac{1}{f'(g(-1))} = \frac{1}{f'(-1)}$$

$$g'(-1) = \frac{1}{4}$$

Hence, tangent is

$$(y+1) = \frac{1}{4}(x+1) \Rightarrow 4y+4 = x+2$$

$$\Rightarrow x-4y=3$$

8. Ans (4)

$$\left(\frac{dy}{dx} \right)_{(x_1, y_1)} = 3x_1^2 + 1$$

$$\Rightarrow 3x_1^2 + 1 = \frac{y_1}{x_1}$$

$$\Rightarrow 3x_1^3 + x_1 = y_1 = x_1^3 + x_1 + 16$$

$$\Rightarrow x_1 = 2, y_1 = 26$$

$$m = \frac{26}{2} = 13$$

Sum of digits = 4

9. Ans (3)

$$x + y = 60$$

$$\Rightarrow x = 60 - y$$

$$\Rightarrow xy^3 = (60 - y)y^3$$

Let $f(y) = (60 - y)y^3$; $y \in (0, 60)$

for maximizing $f(y)$ let us find critical points

$$f'(y) = 3y^2(60 - y) - y^3 = 0$$

$$f'(y) = y^2(180 - 4y) = 0$$

$$\Rightarrow f'(45+) < 0 \text{ and } f'(45-) > 0.$$

Hence local maxima at $y = 45$

$$\text{So } x = 14 \text{ and } y = 45$$

10. Ans (3)

$$|\ln x| = px$$

It is sufficient to find values of p for which $y = |\ln x|$ and $y = px$ has three points in common.

If $y = px$ is passing through points O, A, B then we obtain three roots.

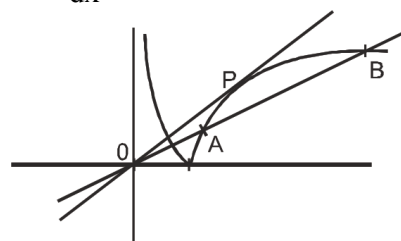
Let us consider line $y = px$. When it is passing through points O, P. (tangent)

Let $P(\alpha, p\alpha)$ ($\alpha > 1$)

$$\Rightarrow p\alpha = |\ln \alpha|$$

$$p\alpha = \ln \alpha \quad \dots(1)$$

$$\frac{dy}{dx} = \frac{1}{x}$$



Figure

$$p = \frac{1}{\alpha} \quad \dots(2)$$

$$(1), (2) \Rightarrow a = e$$

Slope of tangent (OP) is $\frac{1}{e}$.

For three roots condition is $0 < p < \frac{1}{e}$.